

Centre of Gravity

Balancing Acts - Detailed Lesson Plan

Subject	Science / Physics	Duration	60-75 Minutes
Topic	Centre of Gravity & Stability	Academy	PLANRS Academy
Strand	Forces & Motion	Format	Classroom + Hands-On

Learning Objectives

1	Define Centre of Gravity (CoG) as the point where all of an object's weight is concentrated.
2	Explain gravity as an invisible downward pull acting on all objects.
3	Distinguish between stable and unstable objects using everyday Indian examples.
4	Understand the relationship between height of CoG and an object's stability.
5	Locate the balancing point (CoG) of simple objects through hands-on activities.
6	Apply knowledge to predict whether objects will topple or remain steady.

Key Vocabulary

Term	Definition	Child-Friendly Meaning
Centre of Gravity	The point through which gravity acts on an object; the average position of all weight.	"Magic Middle" - the secret spot that keeps an object balanced!
Gravity	The force that pulls all objects with mass toward the Earth.	An invisible hand that pulls everything downward.
Stable	An object that resists tipping and returns to its original position when slightly disturbed.	Strong and steady - hard to push over!
Unstable	An object that topples easily when disturbed; high CoG, narrow base.	Wobbly and easy to tip - falls with a little push!
Balancing Point	The exact point at which an object balances perfectly.	The magic spot on a ruler where it stays still on your finger.
Base of Support	The area beneath an object that supports its weight.	The bottom footprint - wider = steadier!

Background Knowledge for Teachers

The **Centre of Gravity (CoG)** is the single point at which the entire weight of an object can be considered to act. For a uniform object (like a ruler) it is at the geometric centre. For non-uniform objects (like a cricket bat) it is closer to the heavier end.

Stability depends on two factors: (1) the height of the CoG above the base, and (2) the width of the base of support. An object tips when the vertical line through its CoG falls *outside* its base. Wide base + low CoG = highly stable. Narrow base + high CoG = unstable.

In the Indian context this is illustrated by women balancing matkas (clay pots) on their heads, tightrope walkers using long balancing poles to lower their effective CoG, and the lattu (spinning top) using rotational (gyroscopic) stability.

Lesson Structure & Timeline			
Phase	Activity	Duration	Strategy
ENGAGE	Riya's Big Question - tightrope walker at the mela	8 min	Hook / Discussion
EXPLORE	Gravity - The Invisible Puller (mango drop demo)	7 min	Demo + Think-Pair-Share
EXPLAIN	The Magic Balancing Point (CoG concept + 3 Indian examples)	10 min	Direct Instruction
EXPLAIN	Stable vs Unstable Objects (peetha stool vs tall tumbler)	8 min	Comparison Lesson
ELABORATE	Activity 1 - Ruler on Finger (find the magic middle)	8 min	Hands-On Inquiry
ELABORATE	Activity 2 - Cardboard on Pencil (non-uniform object balancing)	8 min	Guided Discovery
ELABORATE	Cricket Bat Balancing Act	5 min	Application
CONNECT	Balance All Around Us (Indian real-life examples)	5 min	Cultural Connection
EVALUATE	Quick Quiz + Balancing Chant	6 min	Formative Assessment
WRAP-UP	Vocabulary Recap + Home Activities	5 min	Consolidation

Detailed Lesson Plan

Phase 1 ENGAGE | Riya's Big Question (8 min)

Teacher Script: Show the opening slide of a tightrope walker at the mela. Ask: "Have you ever seen someone walk on a tightrope? How do they stay up without falling?" Allow 2-3 responses, then pose Riya's question: "What is the secret of balancing?" Write it on the board.

Discussion Prompt: "Think of something you have balanced before - a book on your head, a spoon on your finger. How did it feel? Where did you need to hold it?"

Phase 2 EXPLORE | Gravity - The Invisible Puller (7 min)

Demonstration: Hold a ball up high and ask: "What will happen when I let go?" Drop it. Ask: "Why did it fall DOWN and not up or sideways?" Introduce **gravity** as the invisible force that pulls everything toward the Earth.

Have students stand and jump. Ask: "What brought you back down?" Relate to the mango slide image. Key message: **Gravity is always working, even when we cannot see it.**

Think-Pair-Share: "Name 3 things in your home that gravity acts on every day."

Phase 3 EXPLAIN | The Magic Balancing Point (10 min)

Concept Introduction: "Every object has a special point called the **Centre of Gravity** - the Magic Middle. This is where all the weight of the object seems to be focused. When we support an object at this point, gravity pulls equally from all sides, so it stays still."

Three Indian Examples to Discuss:

Matka on Head	Lattu (Spinning Top)	Ruler on Finger
The round base of the matka rests balanced on the flat surface of the head. The CoG is directly above the support point.	A lattu spins on its tiny tip! Rotation keeps its CoG stable above the point. Without spinning, it would fall immediately.	Slide your finger under a ruler until it balances. That exact spot is the CoG - usually right in the middle for a uniform ruler!

Phase 4

EXPLAIN | Stable vs Unstable Objects (8 min)

Key Principle: Stability depends on (a) how high the CoG is from the ground, and (b) how wide the base of support is.

STABLE OBJECTS	UNSTABLE OBJECTS
Wide Peetha Stool - Sits low and wide. CoG is low. Rule: Wide and low = Strong and stable!	Tall Tumbler - Skinny and tall. CoG is high up. Rule: Tall and thin = Easy to tip!
Full Sandbag - Heavy at the bottom, CoG very close to the floor.	Empty Bottle - Light, tall, narrow base - wobbly and easily knocked over.
Steel Dabba Stack - When aligned carefully, combined CoG stays over the base.	Ganesh Idol (certain styles) - Narrow base, weight higher up. Must be placed carefully!

Hands-On Activities

Activity 1
Ruler on
Finger
(8 min)

Materials: 30 cm ruler per student (or pair)

Steps:

1. Hold ruler horizontally with one finger underneath near one end.
2. Slowly slide your finger toward the centre - do not grip, let it slide naturally.
3. The ruler balances perfectly when your finger reaches the CoG (usually the 15 cm mark).
4. Mark this spot with a pencil.

Key Question: What happens if you attach a rubber band to one end before balancing? Where does the new balancing point shift? (*Answer: toward the heavier end*)

Activity 2
Cardboard
on
Pencil (8
min)

Materials: Cardboard strip (~20 cm x 3 cm), pencil, small eraser or coin

Steps:

1. Try to balance a plain cardboard strip on the vertical tip of a pencil.
2. Stick a small eraser or coin to one end with tape.
3. Try again and find the new balancing point by sliding left and right.

Investigation Question: "Is the balancing point always in the middle? Why or why not?"

Expected Answer: No - when one end is heavier the CoG shifts toward that end.

**Cricket Bat
Balancing
Act
(5 min)**

Discussion (or demo with a bat if available):

Ask: *"Where is the cricket bat's CoG - near the handle or near the blade?"*

Answer: Closer to the **blade**, because that is the heavier part.

Two positions to compare:

- Bat held by the **handle**: heavy blade is up high - **less stable**, harder to control.
- Balanced near the **blade end**: weight is lower - **more stable**.

This is why cricketers learn specific grips - to manage the bat's CoG for control and power!

Cultural Connection - Balance All Around Us

Help students connect science to their cultural experiences. Ask them to share examples from their own communities:

Chai Tray Vendor	Street vendors balance large trays of chai glasses on one flat hand. Muscle memory keeps the tray (and arm) as a stable system.
Bharatanatyam Dancer	Dancers frequently stand on one leg. They lower their CoG by bending knees and extend arms to shift the balance point over their single foot.
Seesaw (Jhula)	The seesaw balances when equal weights sit equidistant from the centre pivot - the fulcrum is at the combined CoG of the balanced system.
Stacking Steel Dabbas	Each lid must be centred over the one below. Off-centre stacking shifts the combined CoG outside the base and the stack topples.
Spinning Lattu	The spinning top uses gyroscopic stability - as it spins fast it resists tipping even on its tiny point.

Why Things Topple Over - The Science

An object topples when the **vertical line from its CoG falls outside the base of support**. Demonstrate with a book: stand it upright (stable), then gradually tilt it. When the CoG is no longer above its base, gravity causes it to fall.

Situation	CoG Position	Result
Book standing normally	Directly above the base	Stable
Book leaning slightly	Still above the base	Still stable
Book leaning too far	Outside the base	Topples!
Matka on head, walking straight	Above the head (base)	Stable
Person tilts too far sideways with matka	Outside the head area	Matka slides!

Riya's Balancing Chant (Mnemonic)

Teach the class this chant. Clap along to the rhythm!

***"Magic middle, gravity down,
Low and wide won't fall around!
High and thin will tip and spin,
Find the middle - let's begin!"***

Formative Assessment - Quick Quiz

Use verbally, as a show-of-hands activity, or as a written exit ticket:

Q1 What do we call the "magic middle" of an object in science?

Answer: Centre of Gravity (CoG)

Q2 Which is MORE stable: a full sandbag or a tall empty bottle? Why?

Answer: Full sandbag - it has a lower CoG and wider base.

Q3 If you add a heavy coin to one end of a ruler, which way does the balancing point move?

Answer: It moves toward the heavier (coin) end.

Q4 Name one example of balancing from Indian daily life.

Answer: Any valid answer: matka on head, chai tray, Bharatanatyam dancer, lattu, etc.

Q5 True or False: Objects with a high CoG are more stable than objects with a low CoG.

Answer: False - a lower CoG means greater stability.

Q6 A tree is blown by wind and tilts. When will it fall over?

Answer: When its CoG falls outside its root base of support.

Home Activities - Be the Balance Master!

Encourage students to explore CoG at home with a parent or sibling:

- * Balance a spoon on the rim of a glass - where is the CoG?
- * Try balancing a fork vs a spoon on your finger - which is easier and why?
- * Find the ruler's balancing point with a grown-up's help. Add a coin - what changes?
- * Draw 3 objects from your home and mark their CoG with a star. Are they stable or unstable?
- * Find 3 stable and 3 unstable objects in your home. Write or draw them in your science journal.
- * Watch a Bharatanatyam video and note when the dancer balances on one leg. What do their arms do?

Differentiation Strategies

Support (Emerging)	Core (On Track)	Extension (Advanced)
- Use physical objects to sort (stable/unstable pile) - Provide a vocabulary card with pictures - Pair with a stronger student for activities	- Complete all core activities - Answer all 6 quiz questions - Draw and label 2 objects with CoG marked	- Research: Why do racing cars have wide, low bodies? - Investigate: How does a tightrope walker's pole help? - Design the most stable possible structure from paper

Materials Required		
Item	Quantity	Purpose
30 cm Rulers	1 per student	Activity 1 - Find the Magic Middle
Cardboard strips (20 cm x 3 cm)	1 per pair	Activity 2 - CoG with non-uniform object
Pencils (for balancing)	1 per pair	Activity 2 - pencil as pivot
Coins or small erasers	2-3 per pair	Activity 2 - adding weight to shift CoG
Small ball (tennis/rubber)	1 (teacher)	Gravity demonstration
Cricket bat (optional)	1 (teacher)	Cricket Bat Balancing Act
Science journals / notebooks	1 per student	Recording observations and drawings
Whiteboard / blackboard	1 (classroom)	Vocabulary, diagrams, and key notes